Q.29 Describe types of column according to load transverse.

Q.30 Explain why load carrying capacity of a short column is more than that of a long column.

Q.31 List the advantages of helical reinforcement. (Any four)

Q.32 Explain why shear reinforcement is not provided in slabs.

Q.33 Compare between one way slab and two way slab. (Any five).

Q.34 Describe the aims of limit state methods.

Q.35 Define column and pedestal with diagram.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Design a simply supported RCC beam of span 3.5m. The beam carries a super imposed load of 400 N/m. Use M20 concrete and Fe 415 steel.

Q.37 A short column 400 mm x 400 mm is reinforced with 4-20 mm diameter bars. Find the ultimate load carrying capacity of the column, if the minimum accentricity is less than 0.05 times the lateral dimensions. The material used are M20 grade concrete and HYSD Fe 415 grade reinforcement.

Q.38 Define pre-stressing. Which are the method of pre-stressing, advantages and disadvantages of pre-stresssed concrete.

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5th Sem / Arch

Sub : Reinforced Cement Concrete (RCC)

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 The unit weight of P.C.C. in KN/m^3 is

a) 20 b) 22

c) 24 d) 26

Q.2 The reinforcement in RCC takes

a) Tensile stresses

b) Compressive stresses

c) Shear stresses

d) Torsional stresses

Q.3 The F.O.S. in working stress method for concrete in direct compression is

a) 1.78 b) 3

c) 4 d) 2

Q.4 The RCC beam in which steel reinforcement is provided only in tension zone is called

a) Singly reinforced b) Double reinforced

c) Over reinforced d) None of above

Q.5 F.O.S. for steel

a) 2 b) 3

c) 1.5 d) 1.78

Q.6 The failure of concrete can occur due to

a) Diagonal tension

b) Diagonal compression

c) Tensile stresses

d) None of them

Q.7 Long column fails by

a) Bending b) Crushing

c) Buckling d) Twisting

Q.8 Buckling in a _______ column is more.

a) Long b) Short

c) Both A & B d) None of them

Q.9 The thickness of a two way slab as compared to one way slab is

a) Less b) Equal

c) More d) None of them

Q.10 Sear reinforcement is not required in

a) Beam b) Column

c) Both A & B d) Slabs

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Plan cement concrete posses very tittle ______ strength .

Q.12 The weight of RCC is ______.

Q.13 Concrete is graded on the basis of ______.

Q.14 The axis where the stresses are zero in a RCC beam is called_____.

Q.15 In an ______ beam section the neutral axis lies below the critical neutral axis.

Q.16 In an ______ beam, the steel reinforcement is provided in the tension zone only.

Q.17 Factor of safety for ______ is 7.8.

Q.18 Diagonal cracks are also known as ______ cracks.

Q.19 The nominal cover is provided to protect the reinforcement bar from ______.

Q.20 The failure of a short column is by ______.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Define reinforced cement concrete with their five uses.

Q.22 Discuss different types of steel reinforcement.

Q.23 Describe why steel is used as a reinforcing materials.

Q.24 Explain types of beam sections with diagram.

Q.25 Define neutral axis with diagram.

Q.26 Differentiate between single reinforced and double reinforced beam (Any five)

Q.27 Explain why nominal cover to reinforcement is provided.

Q.28 Differentiate between long column and short column .(any five).